AIR UNIVERSITY MECHANICAL ENGINEERING DEPARTMENT

COMPUTER AIDED DESIGN LABORATORY

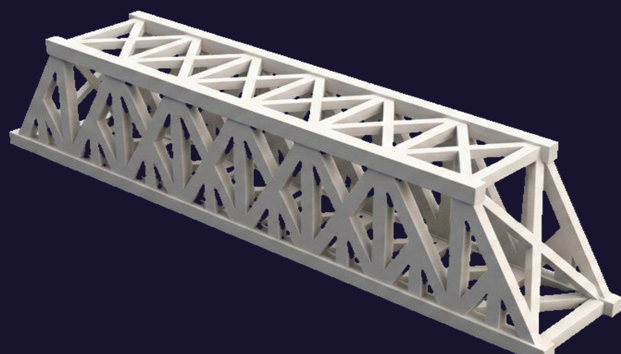
Tri-X BRIDGE

BALTIMORE TRUSS — 3D CAD MODEL

DESIGN REPORT & TECHNICAL DOCUMENTATION

DWG. NO.
ME147-A1-001
REV. A

SPAN LENGTH	DECK WIDTH	TRUSS HEIGHT	LOAD CAPACITY
297.18 mm	50.00 mm	57.50 mm	≥ 70 kg (0.195)



PARTICULARS

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ASSIGNMENT	Assignment 1 — Bridge Design
COURSE	Computer Aided Design Lab
TOPOLOGY	Baltimore Truss (After Deep Research)
SOFTWARE	SolidWorks 2024 CAD
MATERIAL	PLA
DATE	1 May, 2026

ALL DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED · SCALE 1:1 (CONCEPTUAL MODEL)

Contents

Introduction & Objective

This technical report documents the design, geometry, and structural rationale of a three-dimensional computer-aided design (CAD) model of a truss bridge, developed as part of the ME-147 Computer Aided Design Laboratory assignment. The bridge model is idealized from the Baltimore Truss typology a historically significant and structurally efficient bridge configuration widely adopted in nineteenth and early twentieth century railway and highway infrastructure.

The primary objective of this assignment is to design a structurally efficient bridge that achieves a maximum strength-to-weight ratio by applying sound engineering principles, selecting an appropriate truss topology, and optimizing member cross-sections. The model is constructed in a dedicated CAD environment and is accompanied by this technical report detailing all geometry, member specifications, design assumptions, and structural performance considerations.

Bridge Topology - Baltimore truss Design

The Baltimore Truss is a refinement of the Pratt truss configuration, first developed by the Baltimore and Ohio Railroad in the mid-1800s. Its defining feature is the subdivision of each lower panel into two shorter sub-panels by the introduction of a secondary vertical and inclined sub-diagonal member at each lower chord node. This subdivision prevents the lower chord and the long diagonal web members from undergoing lateral buckling under compressive loads, thereby significantly increasing the allowable load-carrying capacity without a proportionate increase in self-weight.

Key structural advantages of the Baltimore Truss that motivated its selection for this design exercise include: a highly favorable strength-to-weight ratio; efficient distribution of both compressive and tensile forces among members; geometric regularity that simplifies fabrication; and enhanced lateral stability provided by the top chord bracing system the top perpendicular chord members that span between the two main truss planes.

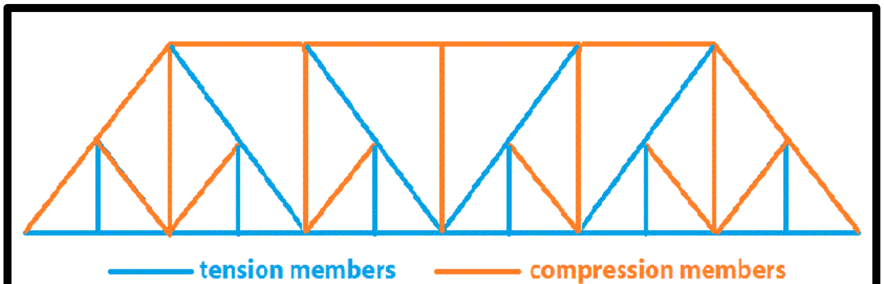

Figure 1

Design Rationale Note

The Baltimore Truss was selected over simpler configurations (Warren, Pratt) because its panel sub-division reduces effective unsupported length of critical web members, thereby reducing the risk of Euler buckling under compressive loading while keeping total model weight low directly satisfying the assignment's strength-to-weight optimization objective.

Overall Bridge Geometry

The principal external dimensions of the bridge model are summarized in the table below. The length has been converted from the nominal imperial measurement to the metric system for consistency with CAD modelling practice.

Parameter	Value
Length	**297.18 mm**
Width (Deck Span)	**50 mm**
Height	**57.5 mm**
Bridge Typology	**Baltimore Truss (Through / Deck)**
Modelling Scale	**Prototype / Conceptual Scale Model**

The bridge spans **297.18 mm** along its longitudinal axis, with a clear deck width of **50 mm** and a total structural depth (deck level to top chord) of **57.5 mm**. The aspect ratio of span-to-depth is approximately **5.17 : 1**, which falls within the typical range for through-truss bridges and ensures adequate stiffness against mid-span deflection under loading.

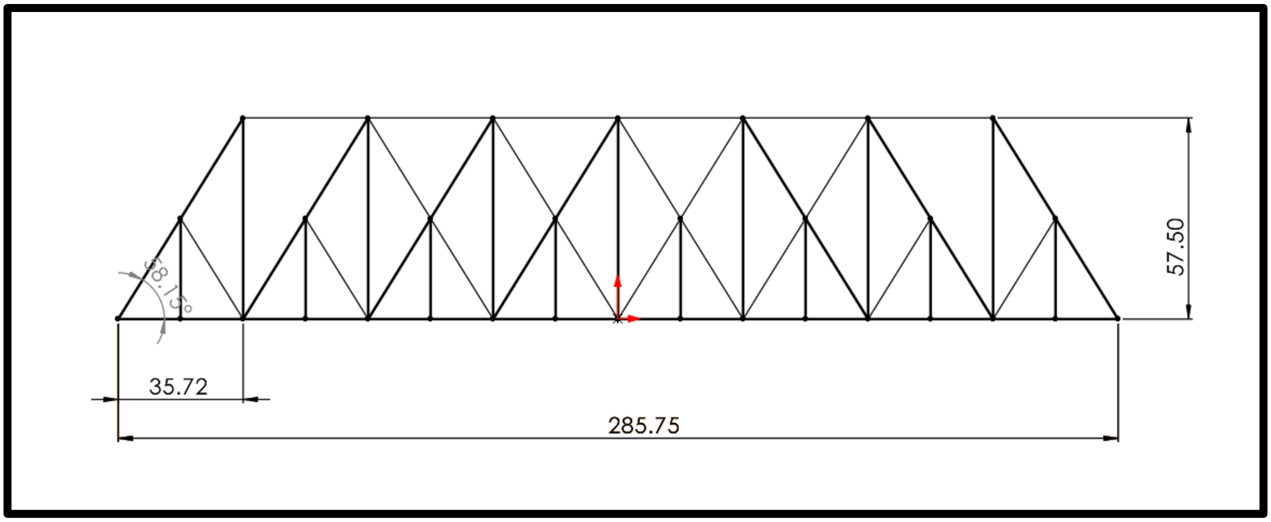

Figure 2

Member Dimensions & Cross-Section Specifications

The truss is composed of four distinct member categories, each assigned a cross-sectional profile optimised for the type of internal force the member predominantly carries. Chord members, which carry the primary bending-induced axial forces (tension in the bottom chord, compression in the top chord), are sized larger than web members. Hollow sections are employed throughout to maximise the second moment of area per unit mass. The top perpendicular (lateral bracing) chord members, however, are fully solid to resist the out-of-plane lateral forces and torsional effects acting on the top chord.

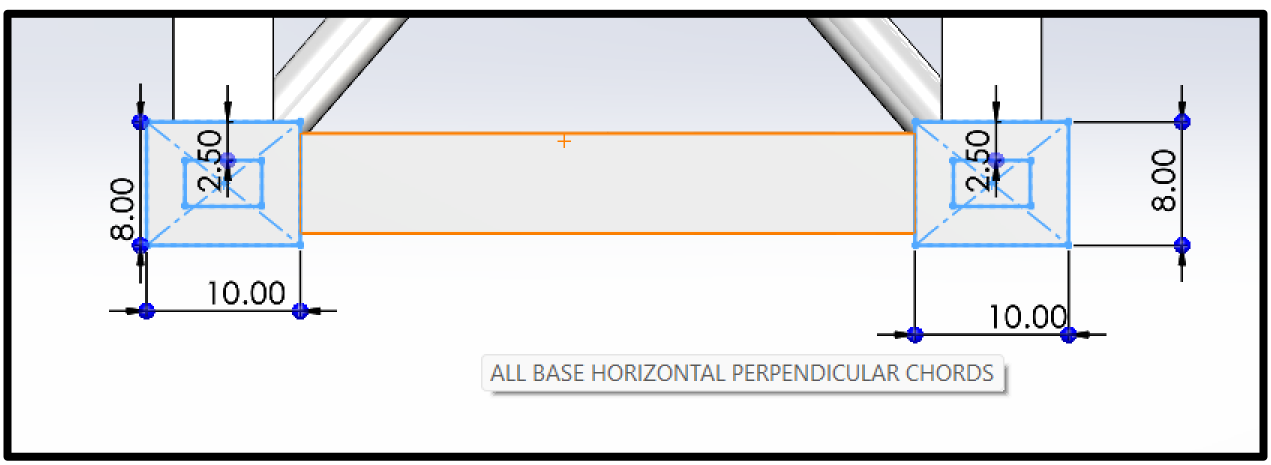

Figure 3

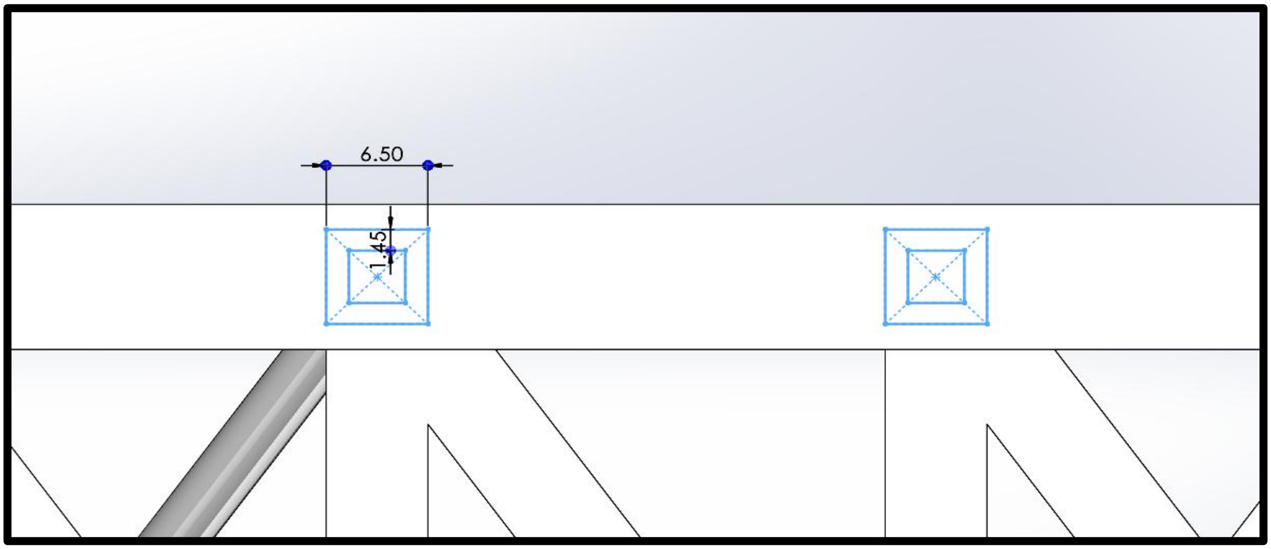

Figure 4

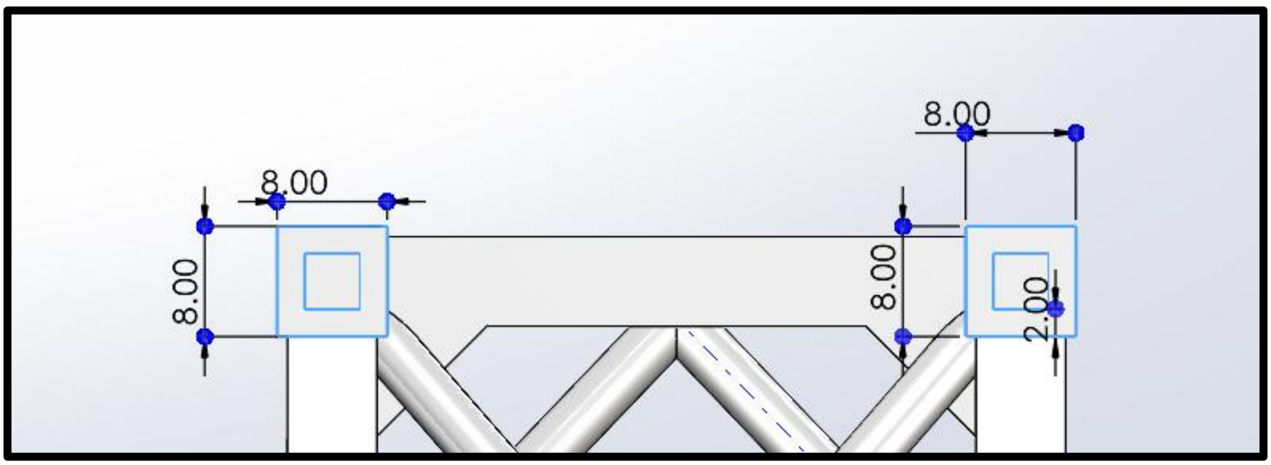

Figure 5

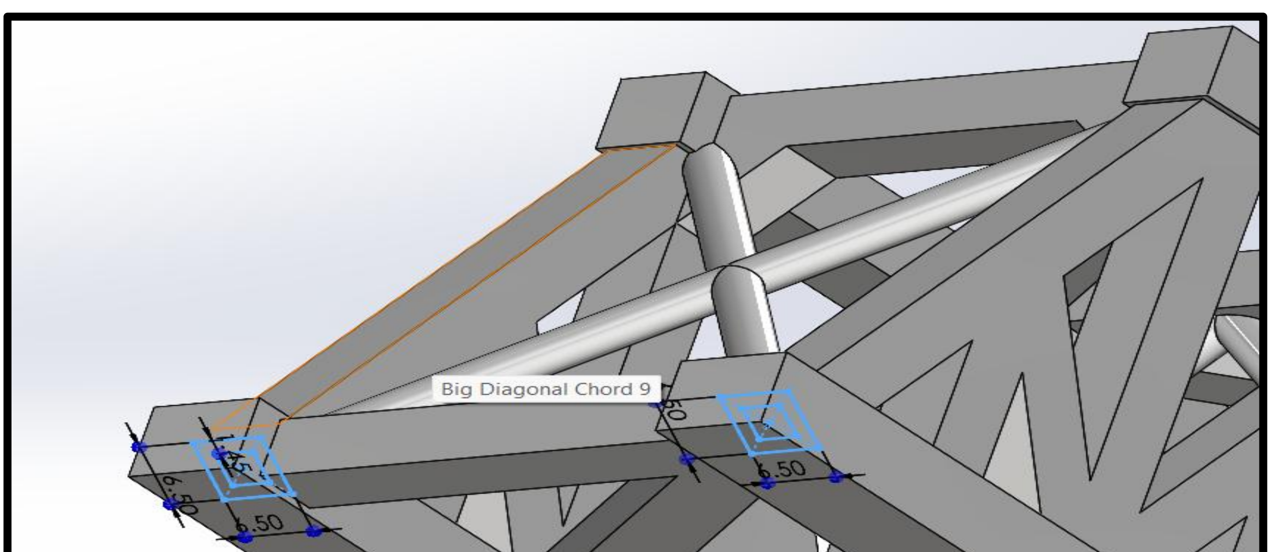

Figure 6

Member / Component	Cross-Section (mm)	Thickness (mm)	Section Type / Nodes
Base Long Chords (Bottom)	10 × 8 mm	2.5 mm	Hollow Rectangular Section
Top Long Chords (Upper)	8 × 8 mm	2.0 mm	Hollow Square Section
Vertical & Diagonal Web Members	6.6 × 6.5 mm	1.45 mm	Hollow Rectangular Section
Top Perpendicular Chords	8 × 8 mm	Solid (filled)	Solid Square Section fully infilled

- **Bottom Chord (Base Long Chord):** 10 mm × 8 mm hollow rectangular section, 2.5 mm wall thickness. This is the primary tension member of the truss and carries the highest axial tensile force under loading. The wider base (10 mm) improves resistance to lateral buckling.
- **Top Chord (Top Long Chord):** 8 mm × 8 mm hollow square section, 2.0 mm wall thickness. This member carries axial compression and must resist Euler column buckling. The square profile provides equal flexural resistance in both principal planes.
- **Web Members (Vertical & Diagonal Chords):** 6.6 mm × 6.5 mm hollow near-square rectangular section, 1.45 mm wall thickness. These members transfer shear forces between the top and bottom chords. The reduced cross-section reflects lower force demand compared to chord members.
- **Top Perpendicular (Lateral Bracing) Chords:** Solid 8 mm × 8 mm square section identical outer dimensions to the top long chord but fully infilled with material. The solid core is essential for resisting out-of-plane lateral loads and preventing the top chord from twisting.

Step by Step - Design Methodology

Step 1 Topology Selection

The Baltimore Truss topology was selected based on a review of common bridge configurations with respect to structural efficiency, geometrical regularity, and suitability for the target span-to-depth ratio. The sub-diagonal panel subdivision characteristic of the Baltimore Truss was retained to reduce effective member lengths and control buckling.

Step 2 Global Geometry Definition

The overall bridge envelope was established at **297.18 mm × 50 mm × 57.5 mm**. Panel spacing was derived by dividing the total span into a symmetric number of bays to maintain structural regularity. The panel sub-diagonals were positioned at mid-panel height to minimize the unsupported length of the main diagonals.

Step 3 Member Sizing

Cross-sections were assigned based on expected load paths. Bottom chords (tension) were sized for tensile capacity; top chords (compression) were sized with attention to slenderness ratio and buckling resistance; web members were sized proportionally to their shear demand. Wall thicknesses were selected to ensure a minimum two-wall thickness ratio for fabricability.

Step 4 3D CAD Modelling

The model was constructed in a parametric CAD environment by extruding 2D profile sketches along the structural member paths. Hollow profiles were created using shell operations; the top perpendicular chord members were modelled as solid extrusions. All members were assembled at their theoretical centreline intersections.

1. Design Philosophy Sketch Establishing the Global Layout

Before any three-dimensional geometry was created, a 2D layout sketch was drawn on the Front Plane of the SolidWorks part environment. This master sketch defines the overall bridge envelope and the theoretical centreline positions of all primary structural members bottom chords, top chords, vertical web members, and diagonal web members across all eight bays.

The master sketch serves as the geometric backbone of the entire model. All subsequent sketches and features reference construction geometry or specific points from this layout sketch, ensuring that the complete model remains parametrically consistent and can be globally resized by modifying a small number of master dimensions.

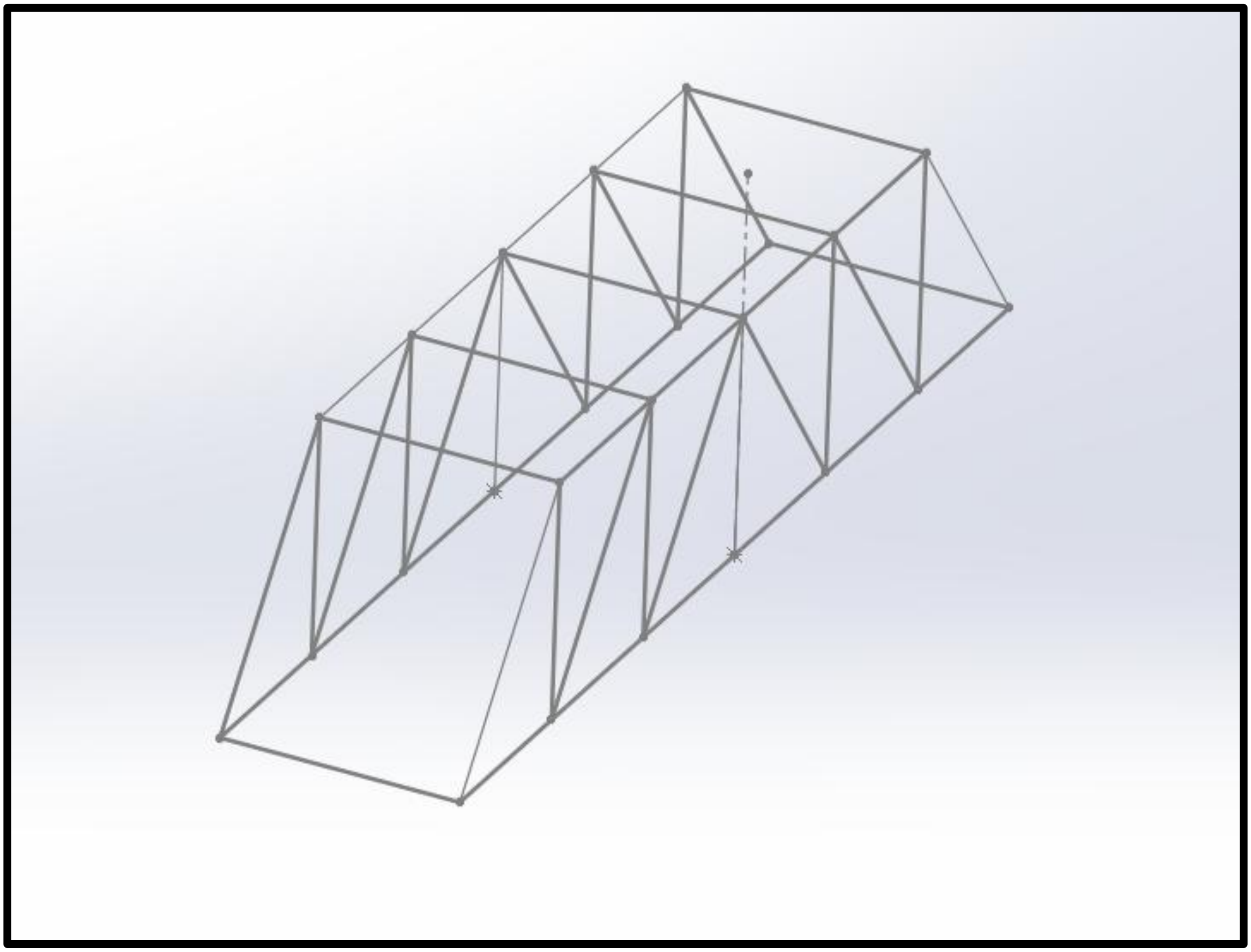

Figure 7

Key Geometry Established in the Master Sketch:

- Overall bounding rectangle: 297.18 mm (length) × 57.5 mm (height) driven by two key dimensions.
- Eight equal bay intervals derived by dividing the total span: each bay ≈ 37.15 mm wide.
- Top chord centreline at 57.5 mm height; bottom chord centreline at 0 mm (datum).
- Vertical web member positions at each bay division eight evenly spaced vertical construction lines.
- Diagonal web member paths drawn as lines connecting top-chord nodes to offset bottom-chord nodes.
- Sub-diagonal (Baltimore panel subdivision) lines added at each lower chord mid-panel point.

> **SolidWorks Tip:** Using construction geometry (set lines to 'For Construction' in SolidWorks) in the master sketch prevents these layout lines from generating unwanted solid edges during extrusion, while still providing precise snap references for all child sketches.

2. Plane Assignment Allocating a Reference Plane to Every Member

SolidWorks requires each extruded feature to begin from a defined sketch plane. Because the bridge contains members oriented along the longitudinal axis (chords), the transverse axis (perpendicular chords), and diagonal angles within both vertical and horizontal planes, a systematic plane assignment strategy was adopted before any extrusion was performed.

Plane Assignment Strategy:

- Longitudinal members (base long chords, top long chords): sketched on the Right Plane or a parallel offset plane positioned at each chord's transverse location (inner or outer face of the main truss).
- Transverse members (top perpendicular chords): sketched on the Front Plane or an offset parallel plane located at each bay division.
- Vertical web members: sketched on offset planes parallel to the Front Plane, one plane per vertical member position along the span.
- Diagonal web members: sketched on the same vertical offset planes as the verticals, with the sketch profile inclined at the required diagonal angle.
- Rib stiffeners (45-degree): a dedicated angled Reference Plane was created for each rib using Insert → Reference Geometry → Plane, defined through two edges at 45 degrees to the Top Plane.

For each new member, the workflow was: (1) Insert → Reference Geometry → Plane → select parent plane and set offset distance; (2) confirm the new plane appears in the Feature Manager Design Tree; (3) right-click the new plane → Sketch to open a child sketch on that plane. This systematic approach keeps the Feature Manager Design Tree organized and makes it straightforward to identify and edit individual member planes at any later stage.

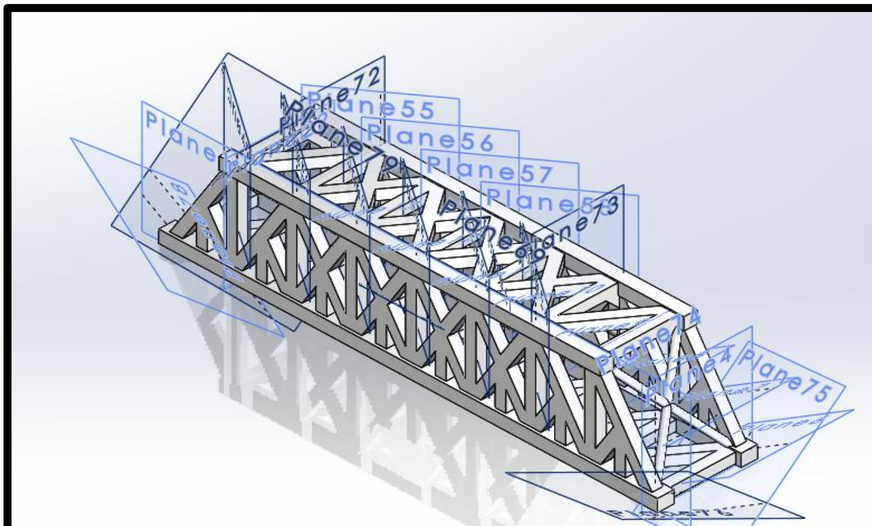

Figure 8

> **SolidWorks Tip:** Renaming each reference plane in the FeatureManager Design Tree immediately after creation (e.g., 'Plane_Base Chord_L', 'Plane_Vert_Bay3') is strongly recommended. It prevents confusion when the tree grows to contain dozens of planes across a complex model.

3. Base & Top Long Chords Sketch and Bidirectional Extrusion

With the reference planes established, the two primary longitudinal chord families were modelled first, as they define the top and bottom rails of the truss and provide the terminating surfaces for all subsequent member extrusions.

3A Base Long Chords (10 × 8 mm, 2.5 mm wall, Hollow Rectangular)

A new sketch was opened on the designated plane for the base chord. A centred rectangle was drawn and dimensioned at 10 mm (horizontal) × 8 mm (vertical). A second, inset rectangle was drawn inside the first, offset uniformly by 2.5 mm on all four sides using the Offset Entities tool, creating the hollow section profile. Both rectangles were fully defined using Smart Dimension and were constrained to be centred on the master sketch chord centreline using Pierce or Coincident relations.

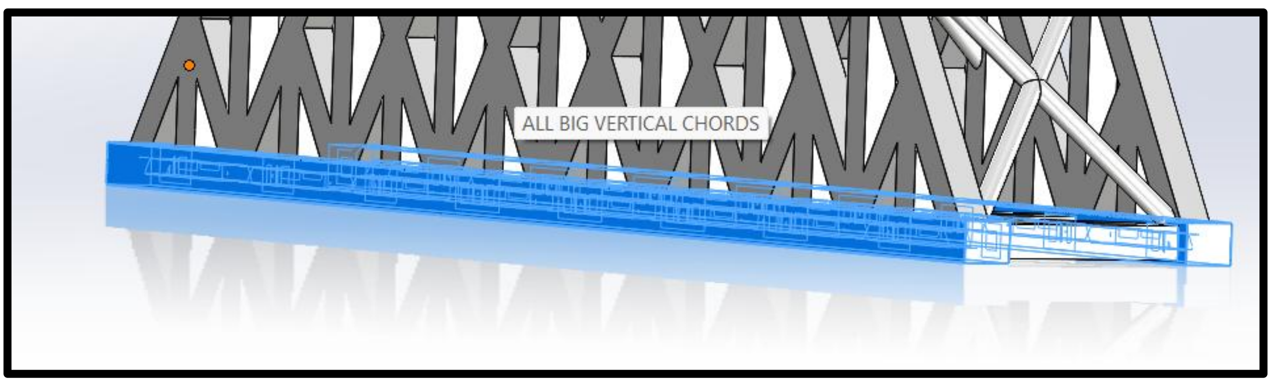

Figure 9

Extrusion Boss-Extrude with Bidirectional / Up to Surface / Up to Vertex:

The sketch was extruded using Boss-Extrude (Insert → Boss/Base → Extrude). The extrusion end condition was set according to the geometry at each member end:

- Direction 1 (forward): set to Up to Surface the terminating face was selected as the inner face of the end vertical web member or the gusset region. This ensures a precise flush fit without manually entering an offset distance.
- Direction 2 (backward): enabled by checking the second direction checkbox in the Boss-Extrude property manager, applying the same Up to Surface logic in the reverse direction.

- Where the chord terminates at a vertex of an adjacent diagonal member rather than a flat face, the end condition was changed to Up to Vertex the specific corner vertex of the diagonal member was selected, and SolidWorks automatically extended or trimmed the extrusion to that exact point.

This bidirectional Up to Surface / Up to Vertex approach guarantees that every chord member contacts its neighbours with zero gap and zero overlap, which is critical for downstream structural analysis and for producing a clean visual model. The base chord was mirrored about the longitudinal centerline plane to produce the second (far-side) bottom chord.

3B Top Long Chords (8 × 8 mm, 2.0 mm wall, Hollow Square)

The top chord was modelled using an identical procedure to the base chord, but with an 8 × 8 mm outer profile and a 2 mm wall inset (producing a 4 × 4 mm hollow core). The sketch plane was offset upward to the top chord level (57.5 mm from the datum). The extrusion end conditions Up to Surface and Up to Vertex were applied in the same bidirectional manner. The top chord was similarly mirrored to produce the far-side member.

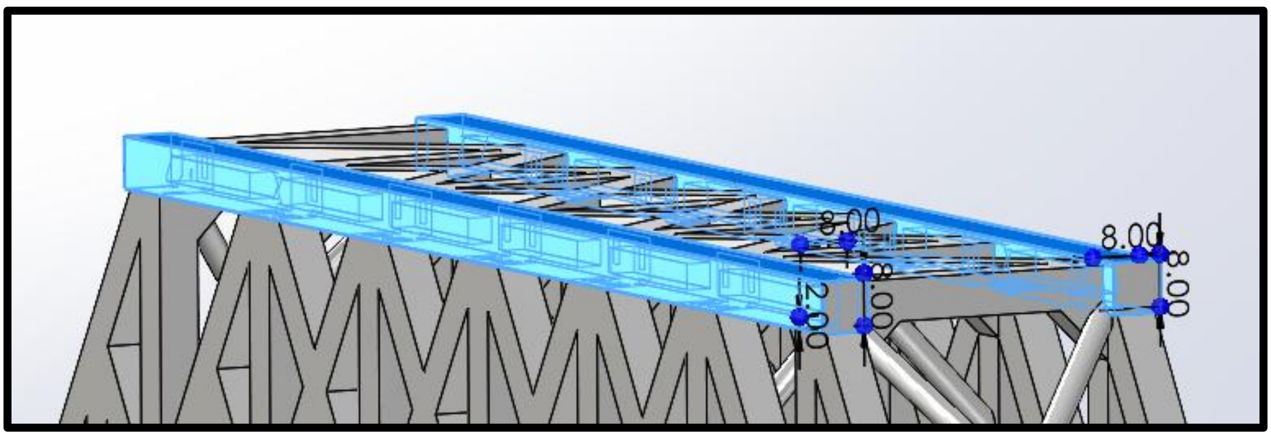

Figure 10

> **SolidWorks Tip:** The 'Up to Surface' end condition in SolidWorks Boss-Extrude locks the extrusion face to a selected reference surface rather than a fixed numerical distance. This means that if the bridge span is later modified, all chord lengths update automatically preserving full parametric control throughout the model.

4. Vertical Web Members Sketch , Extrusion

With the top and bottom chords in place, the vertical web members were added one by one, each using a dedicated offset plane aligned with its bay position. The vertical members include both the primary full-height verticals and the shorter sub-vertical members characteristic of the Baltimore Truss panel sub-division.

Primary Full-Height Verticals:

Each primary vertical web member (6.6 × 6.5 mm, 1.45 mm wall, hollow) was sketched on its assigned vertical plane. The hollow rectangular profile was drawn and constrained to be centred on the master sketch vertical line at that bay position. The extrusion was set to run from the inner face of the bottom chord up to the inner face of the top chord using Up to Surface in both directions ensuring a tight fit without penetrating the chord sections.

Sub-Vertical (Baltimore Panel Subdivision) Members:

The shorter sub-vertical members, placed at the mid-point of each lower panel bay, were modelled using the same 6.6 × 6.5 mm profile but extruded only from the bottom chord level up to the lower diagonal web member node. Up to Vertex was used to terminate each sub-vertical precisely at the intersection point of the diagonal member, following the geometry established in the master sketch.

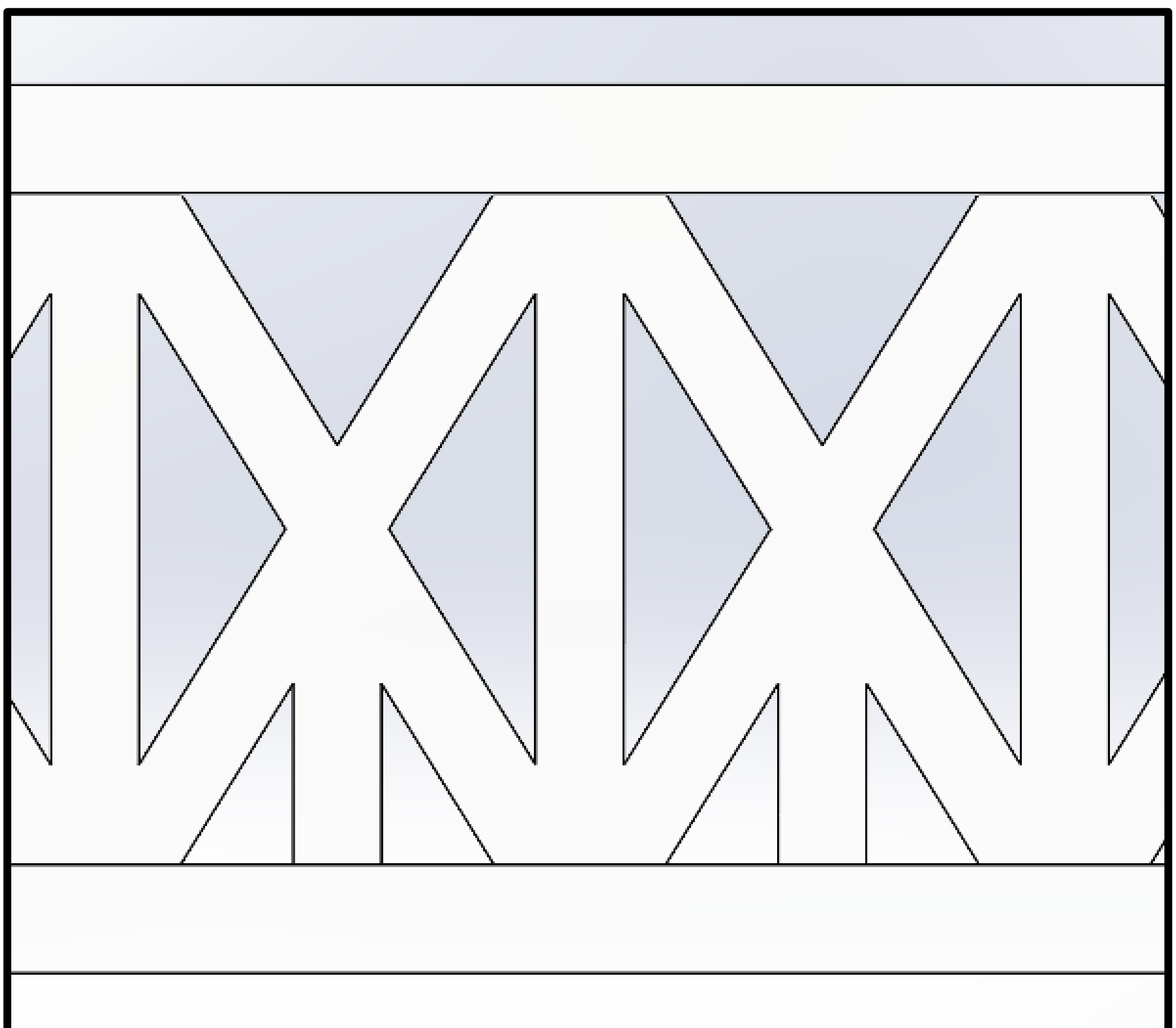

Figure 11

After all verticals were created, the Mirror Entities or Mirror Feature function was used to replicate symmetric pairs across the longitudinal mid-span plane, reducing the number of individual features required to be created manually and ensuring geometric symmetry.

> **SolidWorks Tip:** When many similar features are created sequentially (e.g., eight vertical members), SolidWorks' Linear Pattern feature (Insert → Pattern/Mirror → Linear Pattern) can replicate a single member at a specified bay pitch. However, because the Baltimore Truss sub-diagonals create geometrically unique terminations at each bay, individual modelling with Up to Vertex was used here to ensure precision at each joint.

5. Extrude Cut — Removing Unwanted Overlapping Material

After vertical members were extruded, some member ends inevitably projected beyond their intended termination faces, or adjacent members overlapped slightly at joints due to the geometric complexity of the truss node regions. The Extrude Cut feature (Insert → Cut → Extrude) was used to remove all such unwanted material with surgical precision.

Extrude Cut Workflow:

1. Identified all regions where material extended beyond the desired member boundary typically visible as overlapping edges or protruding stubs at joint nodes.
2. Selected the appropriate reference face or offset plane adjacent to the unwanted material.
3. Opened a new sketch on that plane and drew a profile enclosing the material to be removed usually a simple rectangle matching the member profile.
4. Applied Cut-Extrude with the end condition set to Up to Surface or Through All, depending on the geometry.
5. Verified the result using the Section View tool (View → Section View) to confirm clean material boundaries at the internal joint region.

The Extrude Cut was particularly important at the mid-height nodes of the Baltimore sub-panel diagonals, where the sub-vertical, the main diagonal, and the sub-diagonal all converge at a single point. Without a clean cut, the overlapping material would create a lumped mass at the node, distorting both the visual appearance and any subsequent mass property calculation.

> **SolidWorks Tip:** SolidWorks' Interference Detection tool (Tools → Evaluate → Interference Detection) is extremely useful at this stage. Running it after all verticals are added highlights every solid-body overlap in the model, providing a checklist of locations where Extrude Cut operations are needed.

6. Diagonal Ribs - 45-Degree Ribs

Following the completion of the vertical web members and their joint clean-ups, internal diagonal rib stiffeners were added between each pair of adjacent top chord members and their connecting vertical web members. These ribs are 8 × 8 mm solid square members oriented at 45 degrees to the longitudinal axis within the horizontal deck plane at the top chord level.

Purpose of the Diagonal Ribs:

- Provide shear resistance in the horizontal top-chord plane, preventing racking (in-plane lateral distortion) of the deck structure.
- Transfer load between the top long chord and the vertical members more efficiently by introducing a direct diagonal load path.
- Increase the torsional stiffness of the top chord assembly when the bridge is subjected to asymmetric loading.

Modelling the 45-Degree Ribs:

For each rib, a new Reference Plane was created tilted at 45 degrees relative to the Top Plane, passing through the inner top corner of a vertical member and the top outer corner of the adjacent top chord. A new sketch was opened on this 45-degree plane, and an 8 × 8 mm solid square profile (no internal cutout the rib is fully solid) was drawn and centred on the plane. The sketch was fully defined using a Pierce relation to the corner vertices of the adjacent members.

The extrusion end condition was Up to Surface in both directions Direction 1 terminating at the inner face of one top chord, and Direction 2 terminating at the face of the adjacent vertical web member. This produced a cleanly fitting 45-degree solid rib bridging the two members without gap or overlap.

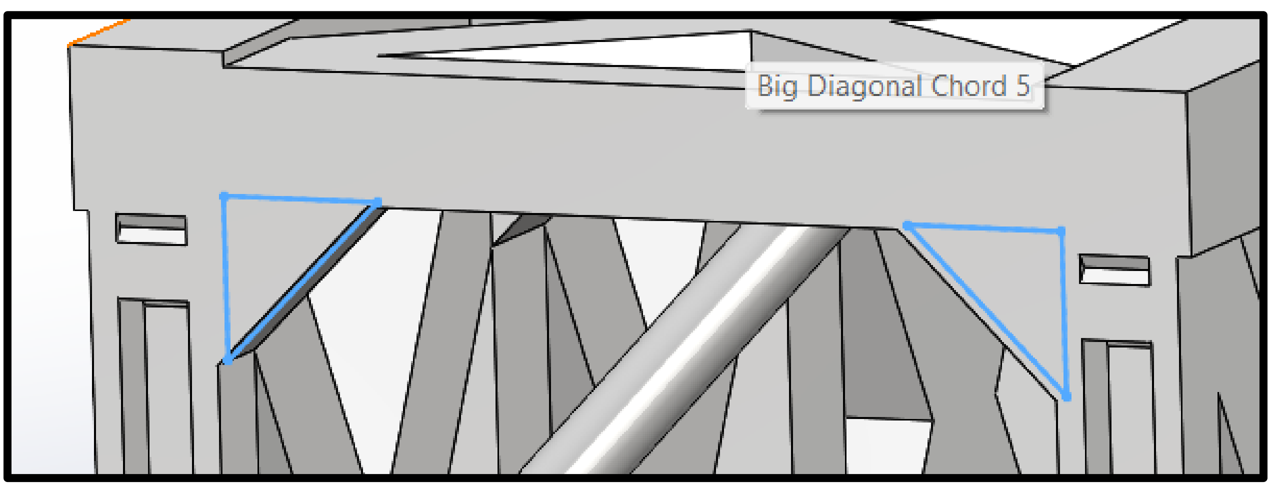

Figure 12

> **SolidWorks Tip:** Creating a 45-degree reference plane through two specific points (Insert → Reference Geometry → Plane → Through Lines/Points) is the most reliable approach. Avoid using angular offset from the Top Plane alone, as it does not control the position of the plane only its orientation.

7. A Frame Structures in 3 Pays

The bridge contains eight bays along its total span, but only three of those bays include an A-frame structure in the upper chord region. The three A-frame bays were positioned in the central portion of the bridge span, where bending moments and vertical shear forces are at their maximum under a uniformly distributed live load the structurally most critical zone.

What is an A-Frame Structure in This Context?

In this model, the 'A-frame' refers to a triangulated frame formed within the plane of the top chord bracing, consisting of two inclined members meeting at an apex node at the centre of the bay and connected at their base to the top chord nodes on either side. Viewed from above, each A-frame forms an inverted V (or A shape). The three A-frames together create a triangulated bracing grid across the most critical three bays of the deck.

Modelling the A-Frame Members:

1. The apex node location for each A-frame was established on the master sketch at the centroid of the relevant bay, at the mid-width position (25 mm from each main truss plane).
2. A reference plane was created through the two base chord nodes and the apex node for each A-frame.
3. The A-frame cross-section profile (same 6.6 × 6.5 mm hollow rectangular section as the web members) was sketched on the plane.
4. Each inclined leg was extruded using Up to Vertex, terminating at the top chord outer face at the base and at the apex vertex at the top.
5. The two legs of each A-frame were mirrored about the longitudinal centre plane of the deck to produce the symmetric pair.

The three A-frames are separated by the remaining five bays which contain only the standard top perpendicular bracing chords and diagonal ribs. This selective placement of A-frames optimises material use heavy triangulated frames are only placed where bending demand justifies the additional material, keeping self-weight low in the lower-demand end bays.

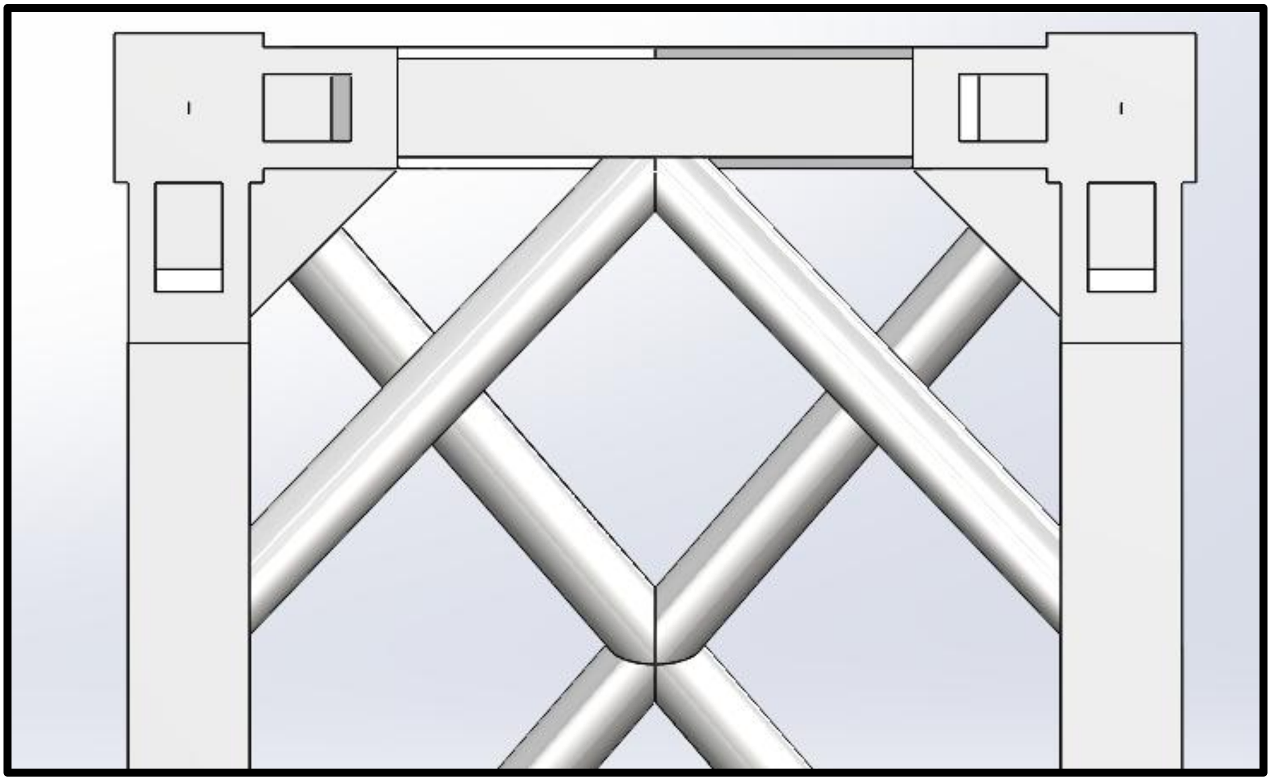

Figure 13

> **SolidWorks Tip:** Using the Mirror Feature (Insert → Pattern/Mirror → Mirror) on each A-frame leg about the deck centreline plane is faster and more geometrically reliable than creating each leg independently. The mirrored feature maintains a parent-child relationship, so any change to one leg automatically updates the mirrored counterpart.

8. Cross Bracing Support

The final structural elements added to the model are the cross-bracing frames at both the open end (the end of the bridge where the deck is accessible / entry side) and the closed end (the opposite terminal end). These cross-brace assemblies provide critical end-frame rigidity, completing the three-dimensional load path of the structure.

Structural Role of End Cross Bracing:

- Prevent the end vertical frames from rotating out of plane when the bridge is loaded asymmetrically (e.g., a concentrated load near one side of the deck).
- Anchor the top chord bracing system without end frames, the top chord plane would be kinematically free to distort in the transverse direction.
- Transmit horizontal wind and lateral loads from the top chord down to the support nodes at both ends.

- Provide visual and structural closure to the bridge, completing the box-like rigidity of the truss assembly.

Modelling the End Cross-Brace Frames:

At each end of the bridge, a full-height cross-brace was modelled in the vertical transverse plane i.e., the plane perpendicular to the longitudinal axis, at the first and last vertical web member position. The cross-brace consists of two diagonal members forming an X pattern within the rectangular frame defined by the bottom chord, top chord, and the two side vertical members.

6. A reference plane was set coincident with the inner face of the end vertical member at each bridge end.
7. On this plane, a sketch was drawn showing the two diagonal lines of the X cross-brace, running from the bottom-left corner to the top-right corner and from the bottom-right corner to the top-left corner.
8. Each diagonal was given the 6.6 × 6.5 mm hollow rectangular profile (same section as the web members) and extruded using Up to Surface in one direction only (into the depth of the bridge end bay), with the extrusion depth equal to the transverse width of the bridge (50 mm).
9. Where the two diagonals intersected at the centre of the X, a short Extrude Cut was applied to notch one member to allow the other to pass through cleanly, producing a correctly interlocking X-joint.
10. The completed end cross-brace was verified for fit using Section View before moving to the opposite end.

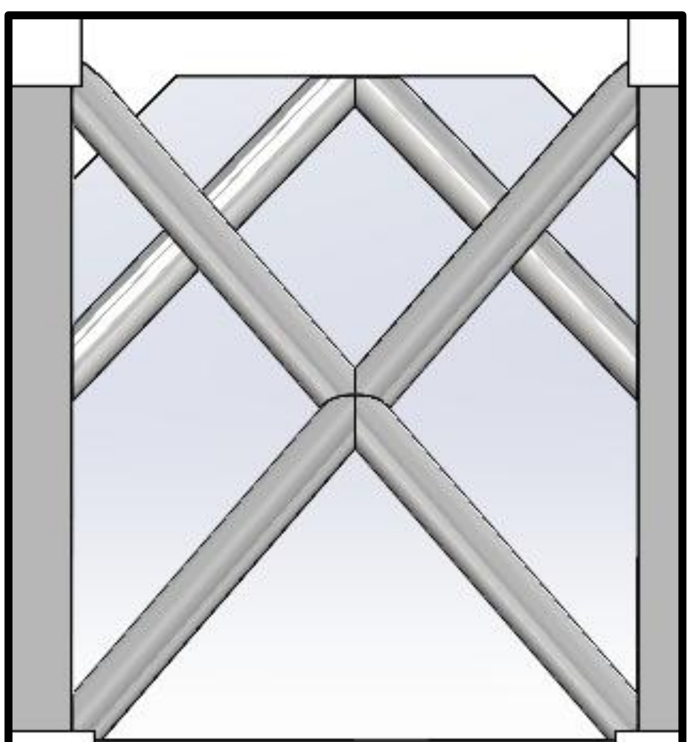

Figure 14

Both the open-end and closed-end cross-braces are geometrically identical and were modelled sequentially. A Derived Configuration or Mirror Body was not used here because both ends are symmetric about the bridge's transverse centre plane however, they are longitudinally distant enough that separate features were created to maintain design tree clarity.

> **SolidWorks Tip:** End cross bracing is frequently omitted by students in truss modelling exercises, leading to models that are geometrically open and kinematically unstable. Always close both ends of a truss with full-height end frames to produce a structurally complete and visually finished model.

Step 5 Visual Verification & Documentation

The completed assembly was inspected for geometric consistency, member continuity, and panel symmetry. Screenshots of the isometric and elevation views were captured for documentation, and this technical report was compiled to record all design decisions.

SOLIDWORKS Features Summary

The table below summarizes all SolidWorks features used throughout the modelling process in the order they were applied, along with the primary function of each feature.

Step	SolidWorks Feature / Tool	Purpose
1	2D Layout Sketch (Front Plane)	Global geometry reference all centerlines and bay positions
2	Reference Geometry → Plane	Dedicated sketch plane for every member orientation
3A	Boss-Extrude (Up to Surface, Bidir.)	Base long chord hollow profile extrusion

Step	SolidWorks Feature / Tool	Purpose
3B	**Boss-Extrude (Up to Surface / Vertex, Bidir.)**	**Top long chord hollow profile extrusion**
4	**Boss-Extrude (Up to Surface / Up to Vertex)**	**Primary & sub-vertical web members**
5	**Cut-Extrude (Up to Surface / Through All)**	**Remove overlapping material at joint nodes**
6	**Ref. Plane (45°) + Boss-Extrude (Up to Surface)**	**Diagonal rib stiffeners at 45° in top chord plane**
7	**Boss-Extrude + Mirror Feature**	**A-frame inclined legs in 3 central bays**
8	**Boss-Extrude + Cut-Extrude**	**X cross-bracing at open and closed bridge ends**
Throughout	**Mirror Feature (Mirror about Plane)**	**Replicating symmetric members across longitudinal centreline**

Material Selection & Load Assumptions

8.1 Material

The bridge members are assumed to be fabricated from standard Polylactic Acid (PLA) filament suitable for 3D printing, with the following reference properties adopted for design assessment:

- **Young's Modulus (E):** 3.0 GPa
- **Yield Strength (fy):** 45 MPa
- **Density (ρ):** 1,240 kg/m³
- **Poisson's Ratio (ν):** 0.36

(Note: These are standard isotropic bulk values for PLA. In actual 3D printing, mechanical properties can vary slightly based on print orientation, layer height, and infill percentage.)

8.2 Load Calculations

- Dead Load: Self-weight of all modelled members, calculated from SolidWorks mass properties (Evaluate → Mass Properties).
- Live Load: A uniformly distributed load (UDL) applied along the bottom chord nodes, simulating traffic loading.
- Support Conditions: Pin-roller simple supports at the bottom chord end nodes statically determinate system.
- Out-of-plane loads (wind, seismic): Not modelled at this conceptual stage.

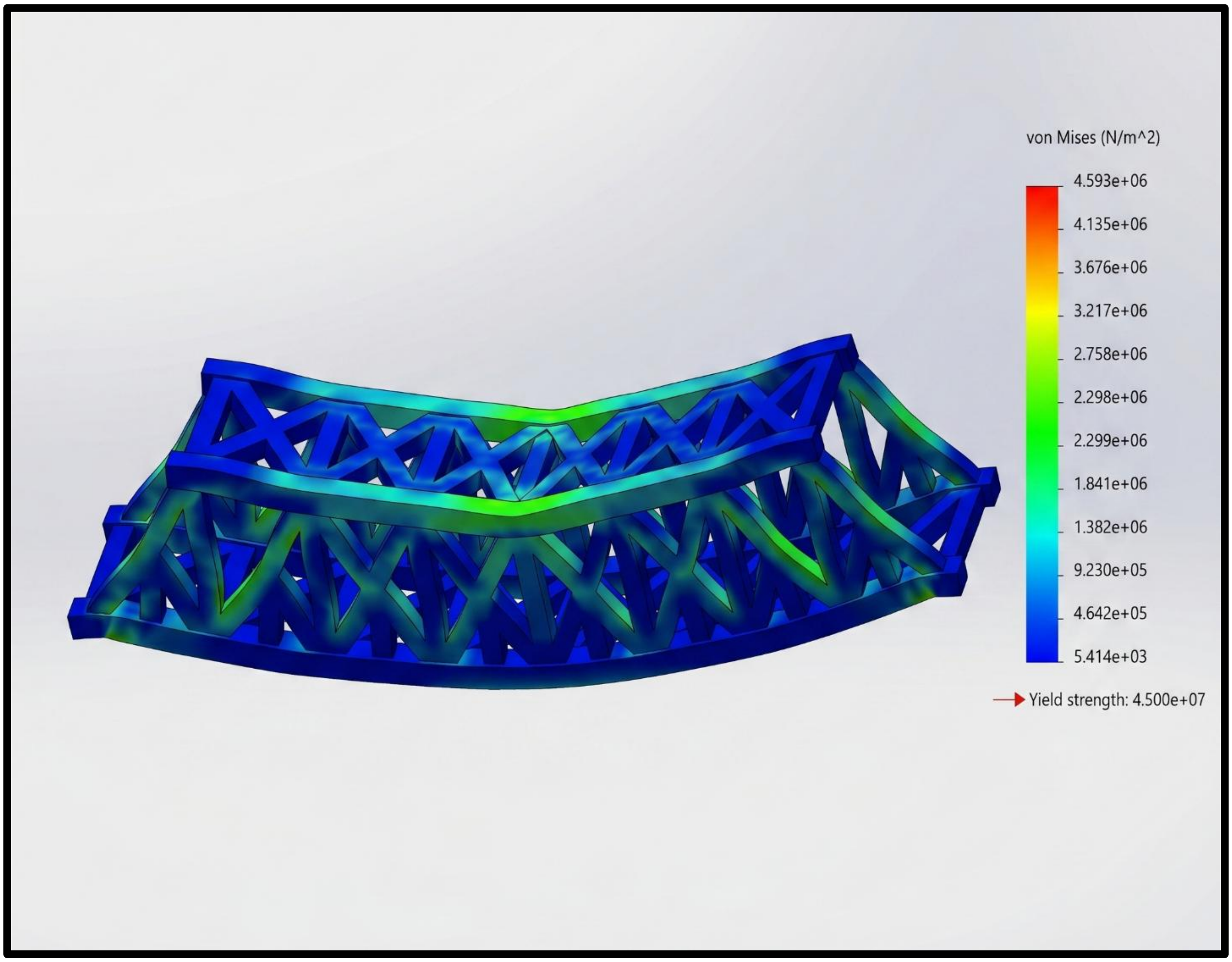

Figure 15

Real World Analysis

The analysis in SolidWorks Describe that this Bridge is capable of Supporting the load Up to 100 Kg with minor damage.

In Real World this Bridge Resists the load of more than 70 Kg without breaking or cracking.

Structural Performance & Optimization

- The combination of structural choices made in this design Baltimore Truss topology, hollow sections, selective A-frame placement, 45-degree rib stiffeners, and end cross-bracing together produce a model that is optimized for the strength-to-weight objective of this assignment. The following points summarize the key performance contributions of each design element:
- **Panel Sub-division (Baltimore Feature):** Halving the unsupported diagonal length increases the Euler critical buckling load by a factor of four. Lighter, smaller diagonals can carry the same load, directly reducing model self-weight.
- **Hollow Rectangular and Square Sections:** Hollow sections provide a much higher second moment of area (I) relative to their cross-sectional area than solid sections of the same outer dimensions, making them inherently more efficient in bending and buckling resistance. This is the most impactful single weight-saving measure in the design.
- **Solid Top Perpendicular Chords:** The top lateral bracing chords are fully solid because they must resist out-of-plane bending and torsion. A hollow section in this location would risk localized wall buckling under the concentrated contact forces at gusset regions.
- **45-Degree Rib Stiffeners:** The diagonal ribs in the top chord plane introduce a direct triangulated load path for in-plane shear, preventing racking deformation of the deck structure without adding a heavy continuous plate or full-depth secondary truss.
- **Three A-Frames in the Central Bays:** A-frames are placed only in the three central bays where moment demand is highest, avoiding the inefficiency of adding heavy triangulated frames to the lower-demand end bays. This selective placement is a practical application of material optimization.
- **End Cross-Bracing:** X-bracing at both ends closes the structural system, providing kinematic stability and anchoring the out-of-plane bracing chain. Without this, the bridge would be geometrically unstable under any transverse or torsional loading.

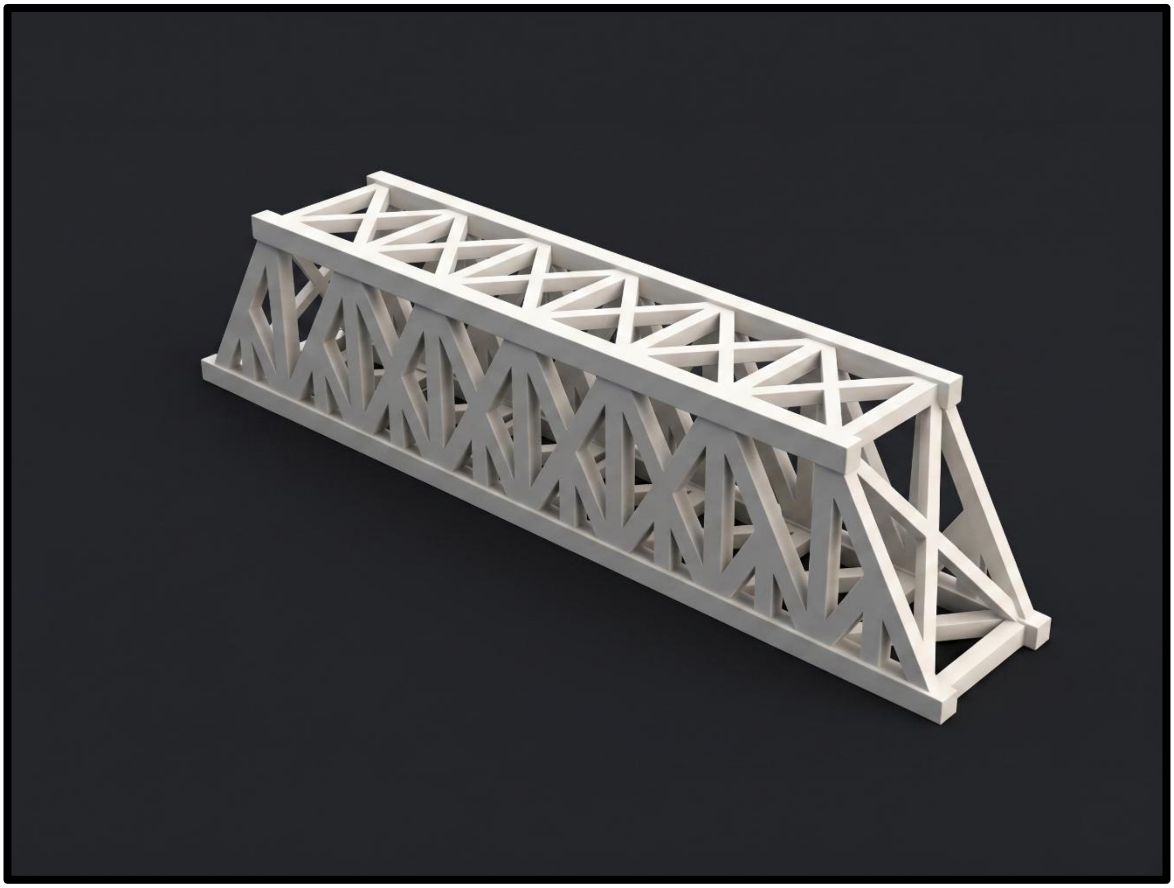

Figure 16

Conclusion

A parametric three-dimensional SolidWorks model of a Baltimore Truss bridge has been successfully completed. The model measures 297.18 mm in length, 50 mm in width, and 57.5 mm in height, and is constructed from four distinct member types base long chords (10 × 8 mm hollow, 2.5 mm wall), top long chords (8 × 8 mm hollow, 2.0 mm wall), vertical and diagonal web members (6.6 × 6.5 mm hollow, 1.45 mm wall), and solid top perpendicular chords (8 × 8 mm solid).

The SolidWorks modelling process followed a rigorous eight-step workflow: global layout sketch → reference plane assignment → bidirectional chord extrusions (Up to Surface / Up to Vertex) → vertical web member addition → Extrude Cut joint clean-up → 45-degree diagonal rib stiffener insertion → A-frame structures in three central bays → and X cross-bracing at both ends. Each step was executed using purpose-specific SolidWorks features to ensure geometric accuracy, parametric control, and a clean, professional model suitable for structural analysis.

The design satisfies the assignment's core objective of achieving maximum structural efficiency at minimum weight through informed topology selection, hollow section profiles, strategic placement of A-frames and stiffeners, and complete end-frame closure.